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ZOO501-Developmental-Biology

Solved Questions FINAL TERM (current PAPERS) February2020

1. Polyspermy 2 marks

The entrance of multiple sperm polyspermy leads to disastrous consequences in most organisms. In the sea urchin, fertilization by two sperm results in a triploid nucleus, in which each chromosome is represented three times rather than twice (Worse condition). Species have evolved ways to prevent the union of more than two haploid nuclei. The most common way is to prevent the entry of more than one sperm into the egg.

polyspermy describes an egg that has been fertilized by more than one sperm. Diploid organisms normally contain two copies of each chromosome, one from each parent. The cell resulting from polyspermy, on the other hand, contains three or more copies of each chromosome—one from the egg and one each from multiple sperm. Usually, the result is an inviable zygote

2. Zona pellucida 2 marks

Zona pellucida: The strong membrane that forms around an ovum as it develops in the ovary. The membrane remains in place during the egg's travel through the fallopian tube. To fertilize the egg, a sperm must penetrate the thinning zona pellucida. If fertilization takes place, the zona pellucida disappears, to permit implantation in the uterus.

The **zona pellucida** supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and free-swimming sperm during and following fertilization.

3. Marble bone disease 2 marks

Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone to breakage (fracture). This disorder is characterized by increased bone density and abnormal bone growth.

Marble bone disease, also called **osteopetrosis** or **Albers-Schönberg disease**, rare disorder in which the bones become extremely dense, hard, and brittle. The disease progresses as long as bone growth continues; the marrow cavities become filled with compact bone. Because increased bone mass crowds the bone marrow, resulting in a reduced amount of marrow and therefore a reduced capacity to produce red blood cells, severe anemia results

4. Bone precursor cell 2 marks

During intramembranous ossification in the skull, neural crest derived mesenchymal cells proliferate and condense into compact nodules. Some of these cells develop into capillaries; others change their shape to become osteoblasts, committed **bone precursor cells.**

What happens when fusion occur between sperm and egg 2 marks

Recognition of sperm by the vitelline envelope or zona pellucida is followed by the lysis of that portion of the envelope or zona in the region of the sperm head by the acrosomal enzymes. This lysis is followed by the fusion of the sperm plasma membrane with the plasma membrane of the egg. The sperm and egg plasma membranes then join together, and material from the sperm membrane can later be found on the egg membrane. The sperm nucleus and tail pass through the resulting cytoplasmic bridge, which is widened by the actin polymerization.

The process of fertilization involves a sperm fusing with an ovum. After the sperm enters the cytoplasm of the oocyte (also called ovocyte), the tail and the outer coating of the sperm disintegrate and the cortical reaction takes place, preventing other sperm from fertilizing the same egg. The oocyte now undergoes its second meiotic division producing the haploid ovum and releasing a polar body. The sperm nucleus then fuses with the ovum, enabling fusion of their genetic material. The successful **fusion** of gametes forms a new organism

5. Fertilin function 2 marks

- In mammals, the fertilin proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion
- Mouse fertilin is localized to the posterior plasma membrane of the sperm head. It adheres the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane

- fertilin appears to bind the sperm plasma membrane to the egg plasma membrane and then to fuse the two of them together. When the membranes are fused, the sperm nucleus, mitochondria, centriole, and flagellum can enter the egg
- Fertilin is a sperm surface protein with an essential role in fertilization. It is required for the migration of spermatozoa through the oviduct, for binding to the zona pellucida, and for efficient binding to the egg plasma membrane

6. Function of dynein 2 marks

The core of the axoneme consists of two central microtubules surrounded by a row of nine doublet microtubules. The force for sperm propulsion is provided by dynein, a protein that is attached to the microtubules. Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.

7. Blastomere progress 2 marks

Blastomeres generally progress through a cell cycle consisting of just two steps:
M (mitosis) and S (DNA synthesis).

8. Embryonic shield 2 marks

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

Also called **embryonic shield**. in the early **embryo** of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the **embryo** begins to differentiate.

9. Example of enhancer in human 2 marks

The human β-globin gene has an enhancer in its 3´ UTR, roughly 700 base pairs

10. What is homology of Hox genes in droshophila and of mammals 2 marks

The homology of gene structure and the similarity of expression patterns between Drosophila and mammalian Hox genes suggests that this patterning mechanism is extremely ancient.

11. Process of neural crest

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

The Neural Crest

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.

The cranial (cephalic) neural crest, whose cells migrate dorsolaterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia,and connective tissues of the face.

Major pathways in differentiation of neural crest

The trunk neural crest, whose cells take one of two major pathways.

- Neural crest cells that become the pigment-synthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.
- The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome.
- Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.
- The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut.
- The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.
- The cardiac neural crest is located between the cranial and trunk neural crests.
- The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

12. Causes in the activity cyclin of MPF

Mitosis-promoting factor contains two subunits.

The large subunit is called cyclin B.
Small subunit of MPF, the cyclin-dependent kinase

Cyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the **cyclin-dependent kinase**.

13. Difference between Enhancer and Promoter?

An **enhancer** is a sequence of DNA that functions to enhance transcription. A **promoter** is a sequence of DNA that initiates the process of transcription. A **promoter** has to be close to the gene that is being transcribed while an **enhancer** does not need to be close to the gene of interest.

14. Mode of commitment

Three basic modes of commitment are,

- ❖ Autonomous specification
- ❖ Conditional specification
- ❖ Syncytial specification

15. What is sea urchin larva name 2 marks

Pluteus larve

16. Cleavage in sea urchin

As the 16-cell embryo cleaves, the eight mesomeres divide to produce two "animal" tiers, an1 and an2, one staggered above the other. The macromeres divide meridionally, forming a tier of eight cells below an2.

17. How neural groove form

Shortly after the neural plate has formed, its edges thicken and move upward to form the neural folds, while a U-shaped neural groove appears in the center of the plate, dividing the future right and left sides of the embryo.

18. Retinoic acid foundation 3 marks

The specification of the ventral neural tube appears to be mediated by external tissues.One agent of ventral specification is the Sonic hedgehog protein, probably originating from the notochord. Another agent specifying the ventral neural cell types is retinoic acid, which probably comes from the adjacent somites.

19. What is hierarchical development 3 marks

The developmental hierarchy is the hierarchical series of decisions involving specific preexisting cell types. These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate. Animal development is the division of embryo into the three germ layers.

1. Ectoderm
2. Mesoderm
3. Endoderm

20. periosteum 3 marks

the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum (a membrane that surrounds the bone).

The **periosteum** is a membrane that covers the outer surface of all bones, except at the joints of long bones. Endosteum lines the inner surface of the medullary cavity of all long bones.

21. Difference between germ cells and somatic cells 5 marks

In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the gametes (the sperm and egg). The gametes and their precursor cells are collectively called germ cells, and they are set aside for reproductive function.

All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development.

22. Difference between fraternal or identical twins 5 marks

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

Identical twins occur in roughly 0.25% of human births. About 33% of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at day 5.

23. Primary neurulation and secondary neurulation 5 marks

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

During primary neurulation, the original ectoderm is divided into three sets of cells.

- ▪ The internally positioned neural tube, which will form the brain and spinal cord.
- ▪ The externally positioned epidermis of the skin.

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary.

Secondary neurulation involves the making of a medullary cord and its subsequent hollowing into a neural tube. In frogs and chicks, secondary neurulation is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae. In both cases, it can be seen as a continuation of gastrulation. The growing region at the tip of the lip is called the chordoneural hinge, and it contains precursors for both the posterior most portion of the neural plate and the posterior portion of the notochord. The tip of the tail is the direct descendant of the dorsal blastopore lip, and the cells lining the blastopore form the neurenteric canal.

24. Three axes of body foundation 5 marks

Three axes are the foundations of the body:

- the anterior-posterior axis (head to tail or mouth to anus),
- the dorsal-ventral axis (back to belly),
- the right-left axis (between the two lateral sides of the body).

25. Step of sperm and egg interaction 5 marks

The interaction of sperm and egg generally proceeds according to five basic step:

➢ The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.
➢ The exocytosis of the acrosomal vesicle to release its enzymes.
➢ The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.
➢ The passing of the sperm through this extracellular envelope.
➢ Fusion of egg and sperm cell plasma membranes.

26. Role of calcium in cortical granule 5 marks

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. The release of calcium from intracellular storage can be monitored visually using calcium activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

27. Structure of sperm 5 marks

Each sperm consists of,

1) A haploid nucleus,
2) A propulsion system to move the nucleus, and
3) A sac of enzymes that enable the nucleus to enter the egg.

During the course of sperm maturation, the haploid nucleus becomes very streamlined, and its DNA becomes tightly compressed

In front of this compressed haploid nucleus lies the acrosomal vesicle, or acrosome, which is derived from the Golgi apparatus and contains enzymes that digest proteins and complex sugars; thus, it can be considered a modified secretory vesicle.

These stored enzymes are used to lyse the outer coverings of the egg. In many species, such as sea urchins, a region of globular actin molecules lies between the nucleus and the acrosomal vesicle.

These proteins are used to extend a fingerlike acrosomal process from the sperm during the early stages of fertilization.

28. Different steps of cleavage in development of snail 10 marks

Spiral holoblastic cleavage is characteristic of several animal groups, including annelid worms, some flatworms, and most molluscs. It differs from radial cleavage in numerous ways..

First, the cleavage planes are not parallel or perpendicular to the animal-vegetal axis of the egg; rather, cleavage is at oblique angles, forming a "spiral" arrangement of daughter blastomeres

Second, the cells touch one another at more places than do those of radially cleaving embryos. In fact, they assume the most thermodynamically stable packing orientation, much like that of adjacent soap bubbles.

Third, spirally cleaving embryos usually undergo fewer divisions before they begin gastrulation, making it possible to follow the fate of each cell of the blastula.

Blastulae produced by radial cleavage have no blastocoel and are called stereoblastulae

The first two cleavages are nearly meridional, producing four large macromeres (labeled A, B, C, and D). In each successive cleavage, each macromere buds off a small micromere at its animal pole

29. Neurlation defects 10 marks

Different neural tube defects are caused when various parts of the neural tube fail to close.

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed. Failure to close the anterior neural tube regions results in a lethal condition, anencephaly. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates. Fetal forebrain development ceases, and the vault of the skull fails to form. The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births

30. Cleavage in mammals 10 marks

The mammalian oocyte is released from the ovary and swept by the fimbriae into the oviduct. Fertilization occurs in the ampulla of the oviduct, a region close to the ovary. Meiosis is completed at this time, and first cleavage begins about a day later. Cleavages in mammalian eggs are among the slowest in the animal kingdom about 12-24 hours apart. Meanwhile, the cilia in the oviduct push the embryo toward the uterus; the first cleavages occur along this journey.

The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction

The cells of the compacted 8-cell embryo divide to produce a 16-cell morula. The morula consists of a small group of internal cells surrounded by a larger group of external cells.

Most of the descendants of the external cells become the trophoblast (trophectoderm) cells. This group of cells produces no embryonic structures. Rather, it forms the tissue of the chorion, the embryonic portion of the placenta.

These cells generate the inner cell mass (ICM), which will give rise to the embryo and its associated yolk sac, allantois, and amnion. By the 64-cell stage, the inner cell mass (approximately 13 cells) and the trophoblast cells have become separate cell layers, neither contributing cells to the other group. During a process called cavitation, the trophoblast cells secrete fluid into the morula to create a blastocoel. The inner cell mass is positioned on one side of the ring of trophoblast cells. The resulting structure, called the blastocyst, is another hallmark of mammalian cleavage.

Once out, the blastocyst can make direct contact with the uterus. The uterine epithelium (endometrium) "catches" the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors

31. What is cleavage 2 marks

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called blastomeres, and by the end of cleavage, they generally form a sphere known as a blastula.

32. Define isolecithal 2 marks

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk")

33. Patterns of embryonic cleavage 10 marks

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. The amount and distribution of yolk protein within the cytoplasm.
2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.
3. The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

34. What is fast block to polyspermy

The Fast block to polyspermy is achieved by changing the electric potential of the egg plasma membrane. This membrane provides a selective barrier between the egg cytoplasm and the outside environment, and the ionic concentration of the egg differs greatly from that of its surroundings. This concentration difference is especially significant for sodium and potassium ions. Seawater has a particularly high sodium ion concentration, whereas the egg cytoplasm contains relatively little sodium. The reverse is the case with potassium ions.

34. Difference between dextral coiling and sinistral coiling.

Some snails have their coils opening to the right of their shells (dextral coiling), whereas other snails have their coils opening to the left (sinistral coiling).

35. Four level of gene expression

The regulation of gene expression can be accomplished at several levels:

Differential gene transcription:

It regulates that which of the nuclear genes are transcribed into RNA.

Selective nuclear RNA processing:

It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

Selective messenger RNA translation:

It regulates that which of the mRNAs in the cytoplasm become translated into proteins.

Differential protein modification:

It regulates that which proteins are allowed to remain or function in the cell.

36. Explain the spina bifida

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed.

37. What is cone growth

Dendrites are often numerous and do not extend far from the neuronal cell body, or soma, axons may extend for several feet.The Nerve outgrowth is led by the tip of the axon, called the growth cone. The growth cone moves by the elongation and contraction of pointed filopodia called Microspikes. These microspikes contain microfilaments, which are oriented parallel to the long axis of the axon.

38. What is hypoblast

Area pellucida cells migrate individually into the subgerminal cavity to form the polyinvagination islands (primary hypoblast). Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

39. Meta cephalon 2 marks

The Metencephalon gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

40. Transcription factor

Transcription factors domains

Transcription factors have three major domains.

1) DNA-binding domain
2) Protein-protein interaction domain
3) Trans-activating domain

DNA-binding domain:

It recognizes a particular DNA sequence.

Protein-protein interaction domain:

It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

Trans-activating domain:

It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this trans-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.

Examples of Transcription factors

MITF:

(Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.

Pax6:

Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains.

Pdx1:

Pancreatic And Duodenal Homeobox 1 gene. It is specific for the pancreatic region of the endoderm.

41. Cortice and nuclei 2 marks

In the cerebellum, some neuronal precursors enter the marginal zone to form clusters of neurons called nuclei

Each nucleus works as a functional unit, serving as a relay station between the outer layers (cortices) of the cerebellum and other parts of the brain

42. Types of cleavage in snail 2 marks

At one extreme are the eggs of sea urchins, mammals, and snails. These eggs have sparse, equally spaced yolk and are thus isolecithal (Greek, "equal yolk"). In these species, cleavage is holoblastic (Greek holos, "complete"). meaning that the cleavage furrow extends through the entire egg.

Zygotes containing large accumulations of yolk undergo meroblastic cleavage, wherein only a portion of the cytoplasm is cleaved. The cleavage furrow does not penetrate into the yolky portion of the cytoplasm.

The eggs of insects have their yolk in the center (i.e., they are centrolecithal), and the divisions of the cytoplasm occur only in the rim of cytoplasm around the periphery of the cell (i.e., superficial cleavage).

The eggs of birds and fishes have only one small area of the egg that is free of yolk (telolecithal eggs), and therefore, the cell divisions occur only in this small disc of cytoplasm, giving rise to the discoidal pattern of cleavage.

43. Events in primary neurulation 5 marks

During primary neurulation, the original ectoderm is divided into three sets of cells.

- The internally positioned neural tube, which will form the brain and spinal cord.
- The externally positioned epidermis of the skin.

The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types

44. What type of molecules involved in histone protein.2 marks

Nucleosome is composed of an octamer of histone proteins (two molecules each of histones H2A-H2B and histones H3-H4) wrapped with two loops containing approximately 140 base pairs of DNA.

45. Difference b/w Karyokinesis and Cytokinesis 5 marks

karyokinesis the mitotic division of the nucleus.

The mechanical agent of this division is the mitotic spindle, with its microtubules composed of tubulin (the same type of protein that makes up the sperm flagellum).

The second process is **cytokinesis** the division of the cell.

The mechanical agent of cytokinesis is a contractile ring of microfilaments made of actin (the same type of protein that extends the egg microvilli and the sperm acrosomal process).

The relationship and coordination between these two systems during cleavage is depicted in Figure, in which a sea urchin egg is shown undergoing first cleavage.

46. meroblastic

The cell divisions do not completely divide the egg, so this type of cleavage is called **meroblastic**.

47. Dose ascidians have notochord 2 marks

Ascidians, members of the tunicate subphylum, are fascinating animals for several reasons, but the foremost is that they are invertebrate chordates. They have a notochord as larvae (and therefore are chordates), but they lack vertebrae.

48. What BHLH protein specific for. 2 marks

the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell.

49. What is promoter 3 marks

Promoter:

A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

50. Define air space 3 marks

An air space forms when the contents of the egg cool and contract after the egg is laid. The air cell usually rests between the outer and inner membranes at the egg's larger end.

51. Describe osteoclast in detail 5 marks

Osteoclasts are probably derived from the same precursors as macrophage blood cells, and they dissolve both the inorganic and the protein portions of the bone matrix.

Each osteoclast extends numerous cellular processes into the matrix and pumps out hydrogen ions onto the surrounding material, thereby acidifying and solubilizing it.

The blood vessels also import the blood-forming cells that will reside in the marrow for the duration of the organism's life. The number and activity of osteoclasts must be tightly regulated.

If there are too many active osteoclasts, too much bone will be dissolved, and osteoporosis will result. Conversely, if not enough osteoblasts are produced, the bones are not hollowed out for the marrow, and osteopetrosis results.

52. Write classes of cadherins 5 marks

In vertebrate embryos, several major cadherin classes have been identified:

1. E-cadherin
2. P-cadherin
3. N-cadherin
4. Protocadherins

E-cadherin: a. Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.

P-cadherin: a. Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.

N-cadherin: a. Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their E-cadherin expression. It is also highly expressed on the cells of the developing central nervous system.

Protocadherins: a. Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

53. Definer silencer 2 marks

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

54. Difference between sex and reproduction 5 marks

Sex and reproduction are two distinct and separable processes. Reproduction involves the creation of new individuals. Sex involves the combining of genes from two different individuals into new arrangements.

Sex and reproduction are two different processes that can or not to occur simultaneously. Sex must be considered as an exchange of genetic material doing new combinations, that can originate or not new individuals. While that the reproduction has that obligatorily to originate new beings.

55. Differentiation process of medulla and spinal chord 10 marks

As the cells adjacent to the lumen continue to divide, the migrating cells form a second layer around the original neural tube. This layer becomes progressively thicker as more cells are added to it from the germinal neuroepithelium. This new layer is called the mantle (or intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma).

The mantle zone cells differentiate into both neurons and glia. The neurons make connections among themselves and send forth axons away from the lumen, thereby creating a cell-poor marginal zone. Eventually, glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the Gray matter; the axonal, marginal layer is often called the White matter. In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

As the neural tube matures, a longitudinal groove the sulcus limitans divides it into dorsal and ventral halves. The dorsal portion receives input from sensory neurons, whereas the ventral portion is involved in effecting various motor functions.